
Request to Schedule Pre-Conference Talk for IEEE GLOBECOM 2026

dkpra@mech.iitkgp.ac.in <dkpra@mech.iitkgp.ac.in>
To: Ayush Debnath <ayush.d@kgpian.iitkgp.ac.in>

Wed, Aug 26, 2026 at 9:15 PM

From: "Nilanjan Das Chakladar" <ndaschakladar@gmail.com>
To: "Faculties at mech" <faculty@mech.iitkgp.ac.in>
Cc: "goutam" <goutam@mech.iitkgp.ac.in>, "Dr. Korak Sarkar" <koraksarkar@mech.iitkgp.ac.in>, "Dr. Poonam Sundriyal" <psundriyal@mech.iitkgp.ac.in>, "dkpra" <dkpra@mech.iitkgp.ac.in>
Sent: Wednesday, August 26, 2026 9:08:46 PM
Subject: Fwd: Request to Schedule Pre-Conference Talk for IEEE GLOBECOM 2026

Dear Colleagues,

A pre-conference seminar is scheduled tomorrow, Thursday (27 Aug 2026) in online mode from 11 to 11:30 am.

Topic of the seminar: **OmniMed-FL: A Robust Multimodal Federated Learning Framework for Clinical Diagnosis**

Abstract:

Simultaneous assessment of medical imaging and patient records is required in clinical diagnosis. But standard machine learning algorithms cannot analyze these several sorts of data together. Meanwhile, severe privacy restrictions such as HIPAA and GDPR also preclude hospitals from aggregating sensitive patient data into a central computer. This leaves a crucial void: secure fusion of visual and textual context across distant networks. We propose OmniMed-FL, a multimodal federated learning framework for multi-class clinical condition classification (Normal, Pneumonia, COVID-19, Pleural Effusion, Cardiomegaly). The framework benchmarks five LLM text encoders, five Vision Transformer (ViT) image encoders, and eight VLM fusion strategies under FedAvg with non-IID Dirichlet($\alpha=1.0$) partitioning across 5 hospital clients. An anti-collapse stack combining balanced batch sampling, entropy diversity regularization, and adaptive early abort maintains full five-class prediction diversity throughout training. DistilBERT achieves an F1 score of 0.934 on text; ViT-Base achieves an F1 score of 0.664 on images; Concat VLM fusion achieves an F1 score of 0.956, a +2.3% gain over the best unimodal baseline. Under strict non-IID conditions, federated training retains 99.1% of centralized accuracy: Fed-VLM matches its centralized counterpart, proving that robust multimodal diagnosis is possible without sharing patient data. A FAISS-indexed RAG pipeline provides contextgrounded retrieval with cosine similarity exceeding 0.89 across all conditions.

Kindly join if available.

Google meet link: <https://meet.google.com/ecw-ixwi-mnu>

BW
Nilanjan Das Chakladar

Nilanjan Das Chakladar (PhD)

Associate Professor

Mechanical Engineering Department

Indian Institute of Technology Kharagpur

Kharagpur, INDIA 721302

<https://sites.google.com/view/nilanjan-das-chakladar/ace-lab-iitkgp>

Fellow: IMechE(UK), IOM3(UK); Professional Member: ASME

Chairperson, CoEAMT
<https://coeamt.iitkgp.ac.in/>
Joint Convenor, AMMES 2027
<https://ammes2027.iitkgp.ac.in/>

----- Forwarded message -----

From: <dkpra@mech.iitkgp.ac.in>
Date: Tue, 25 Aug 2026 at 22:39
Subject: Fwd: Request to Schedule Pre-Conference Talk for IEEE GLOBECOM 2026
To: goutam <goutam@mech.iitkgp.ac.in>, Dr Nilanjan Das Chakladar
<ndaschakladar@mech.iitkgp.ac.in>
Cc: Ayush Debnath <ayush.d@kgpian.iitkgp.ac.in>

[Quoted text hidden]

[Quoted text hidden]

3 attachments



Adobe Scan 25-Aug-2026.pdf
82K



1571289364 paper.pdf
1025K



Indian Institute of Technology Kharagpur Mail - [IEEE GLOBECOM 2026] Your paper #1571254564 ('6G Network empowering AI agents').pdf
145K